S&P 500 Triplet-Factor Model: A Systematic Approach to Momentum-Volume Trading

2026.3.17

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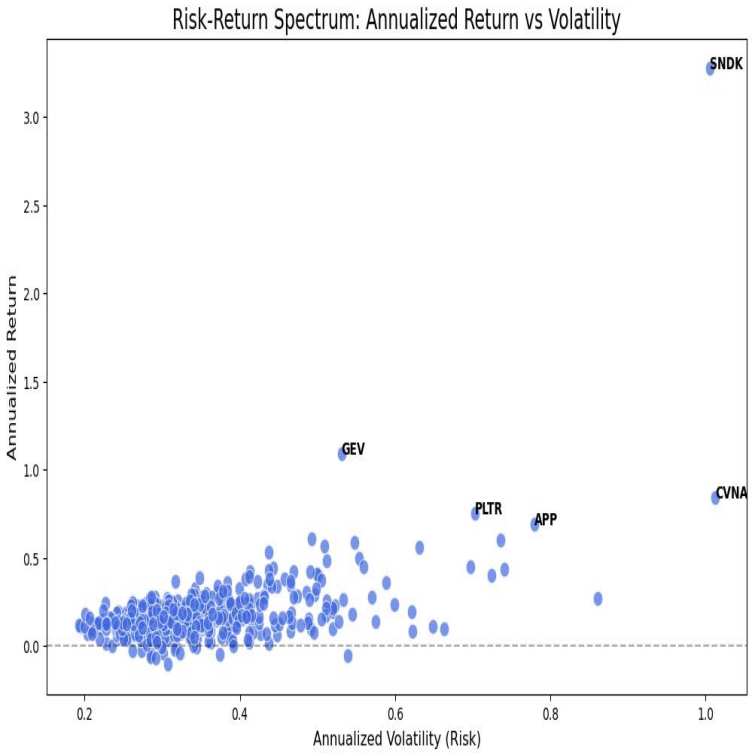
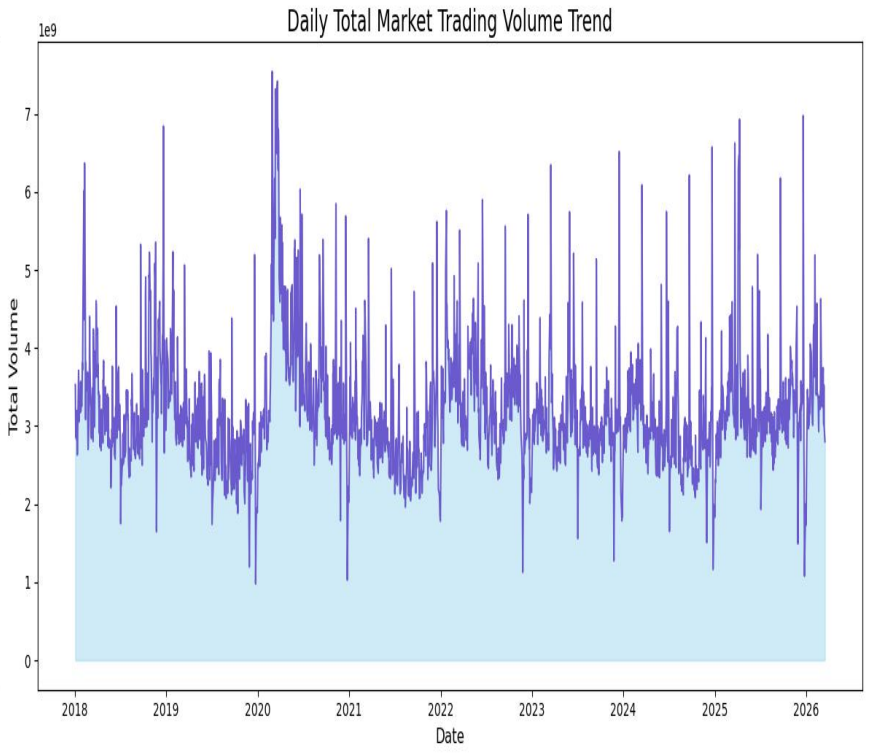
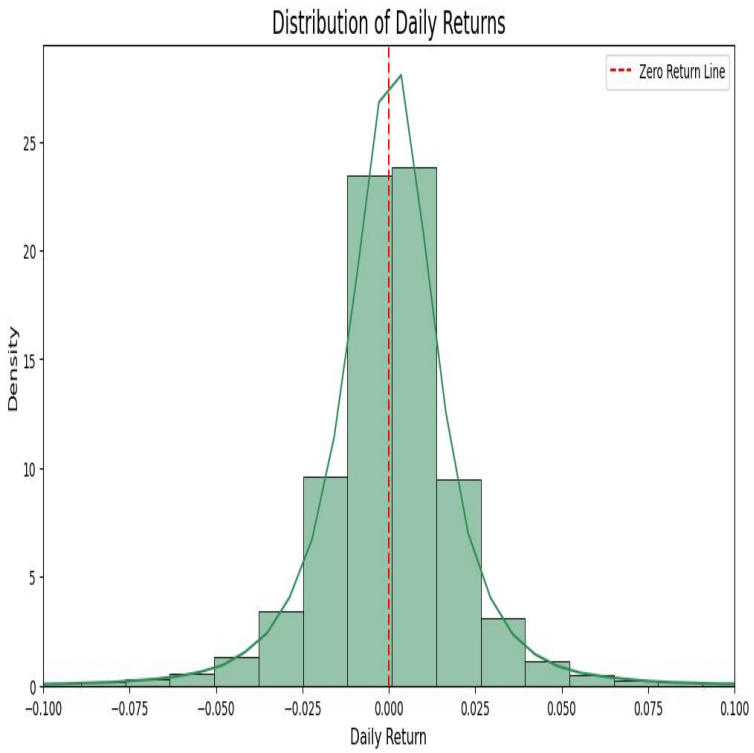
01.

Data Description







Risk-Reward:
A distinct cluster exists in the bottom-left, representing low-volatility, index-like performance.

The Cluster:
Most assets show low-yield stagnation (bottom-left).

Cycle Coverage:
2018–2026 data captures bull, bear, and "Black Swan" regimes.

Volatility Spikes:
Volume surges (e.g., 2020) pinpoint macro stress periods.

Execution Safety:
Multi-billion daily volumes ensure high capacity and low slippage.

Zero-Mean:
Returns cluster at zero, reflecting baseline efficiency.

Risk Mitigation:
The presence of "Fat Tails" necessitates robust risk controls and portfolio diversification to defend against Black Swan shocks.

02.

Factor Logic

Qualitative Analysis: Economic Intuition

(Mom_Gap)

Logic:

Captures the "Information Shock" occurring while the market is closed.

Persistent opening gaps indicate strong overnight sentiment and information-driven momentum that isn't immediately fully priced in during the previous day’s session.

Rel_Strength

Logic:

Measures cross-sectional outperformance relative to the S&P 500 average.

It identifies "Winners" that exhibit persistent alpha. By subtracting the market mean, we isolate stock-specific strength from general market beta.

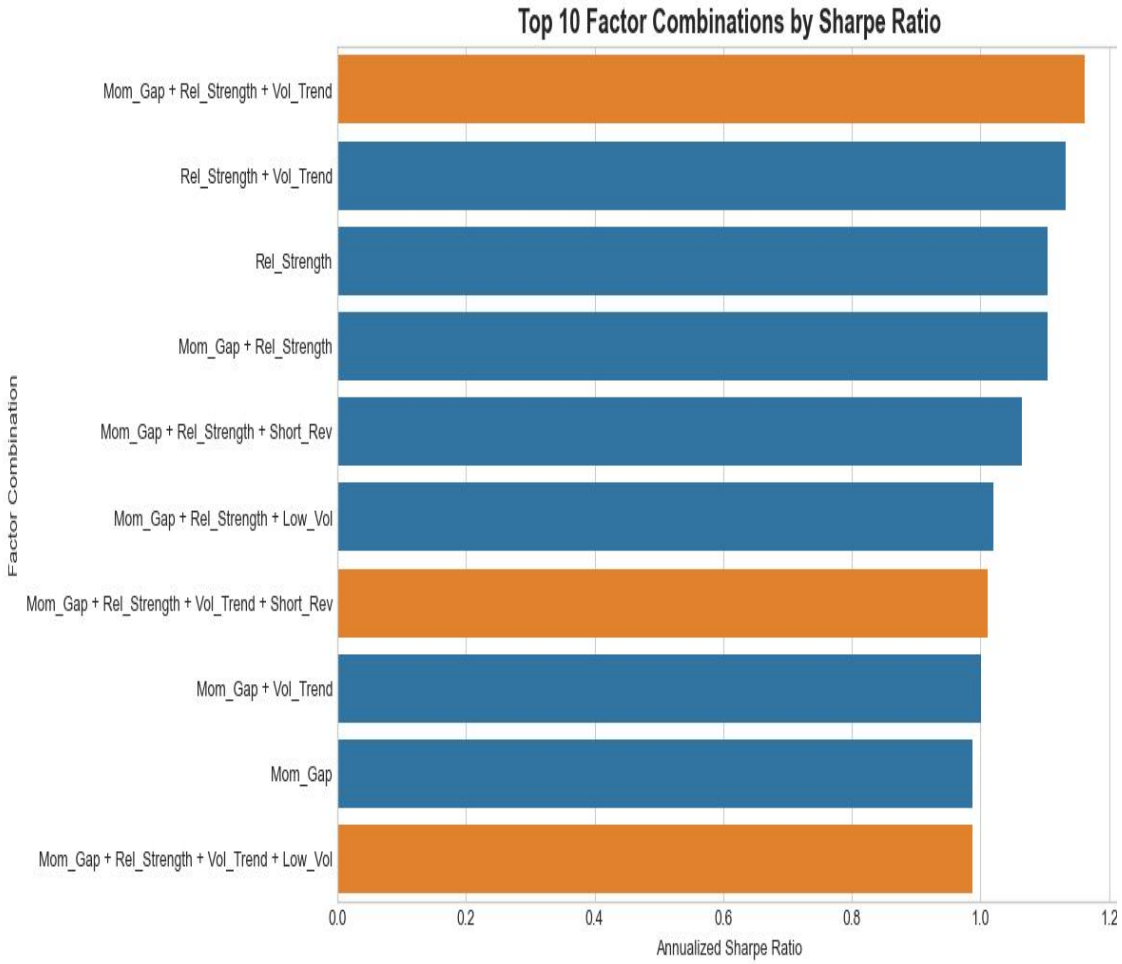
Vol_Trend

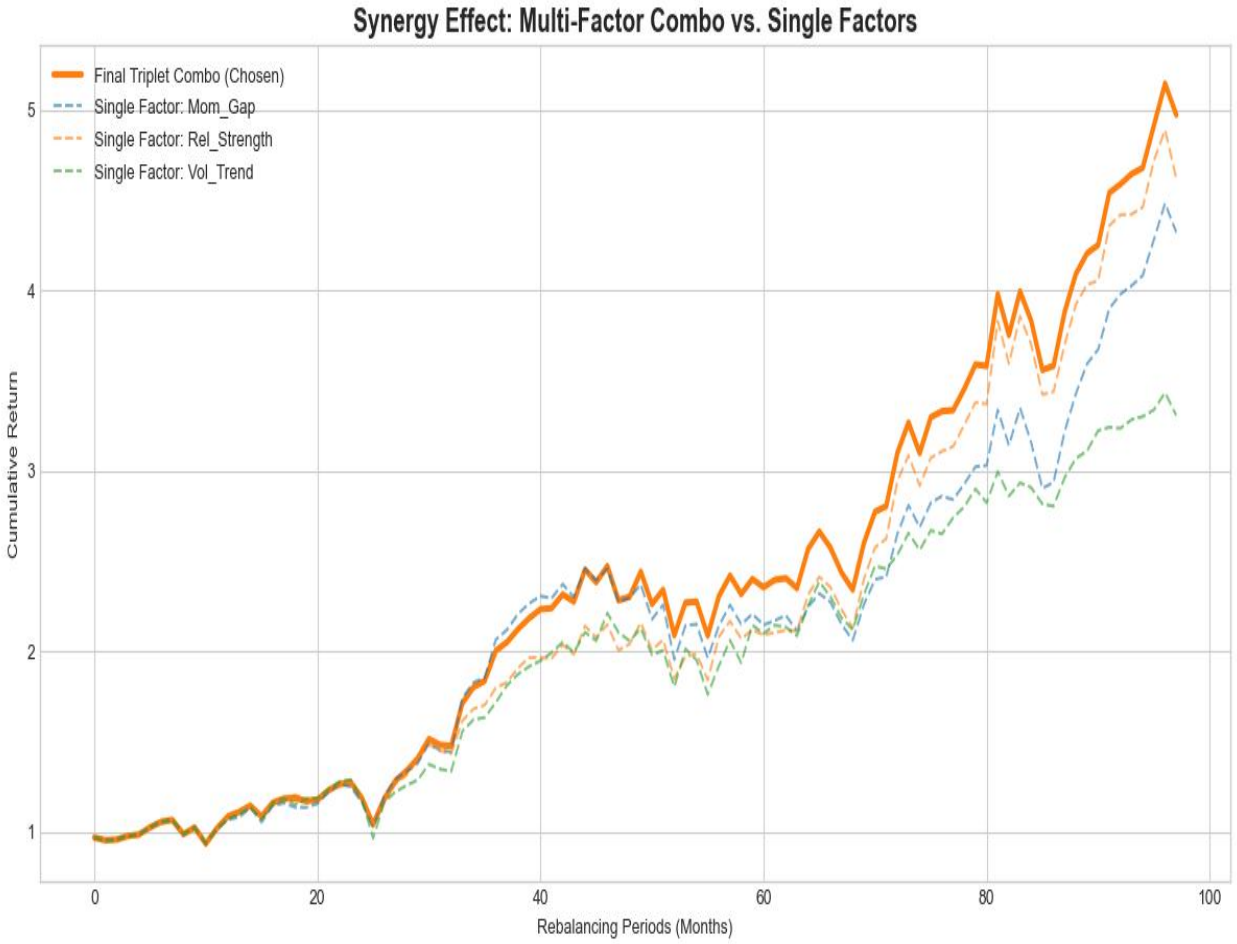
Logic:

Short-term (20D) vs. Long-term (252D) volume ratio.

Volume acts as the "Fuel" for price moves. A surge in relative volume suggests institutional accumulation or a high-conviction trend breakout.

Quantitative Analysis: The Screening Process





Systematic Traversal: Evaluated 31 combinations to identify the mathematically superior signal.

Top Performer: The Triplet Combo (Gap + Strength + Vol) achieved the highest Sharpe Ratio (~1.18).

Stability: This combination consistently outperformed single-factor models across the 2018–2026 horizon.

Diversification: The Triplet Combo (Orange) provides a smoother equity curve by offsetting individual factor drawdowns.

1+1+1 > 3: Combining Sentiment, Momentum, and Liquidity captures a more robust Alpha profile.

Risk Mitigation: The multi-factor approach significantly reduces volatility compared to "pure" price-only strategies.

03.

Strategy Design

Construction and backtesting of multi factor quantitative stock selection strategy

Strategy Overview: Integrating three major factor, through **monthly dynamic position adjustment**, screening potential excess return stocks, and constructing an investment portfolio. Calculate and adjust position at the **end of each month**.

Signal generation and execution:

• Standardization and synthesis: After factor standardization processing and equal weight averaging, generate the final stock selection signal.

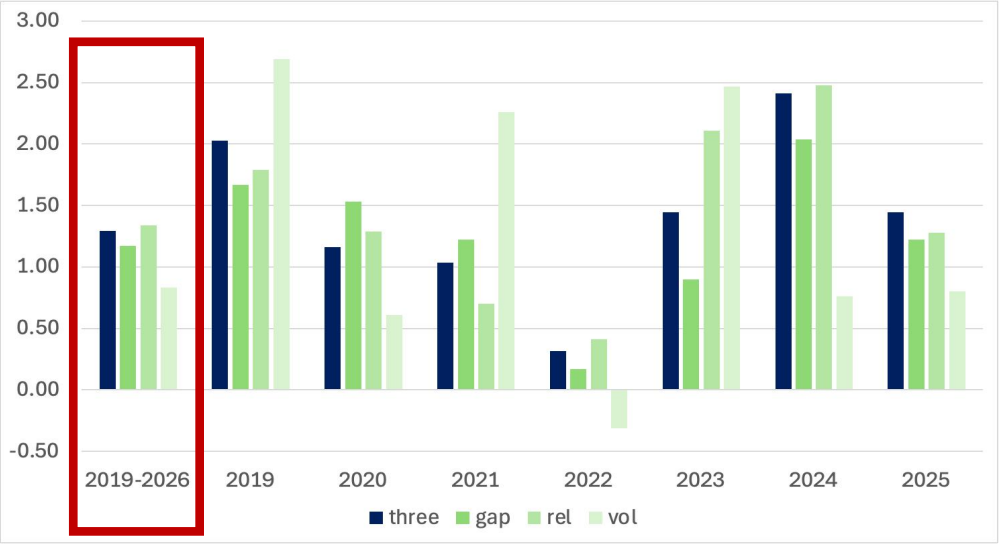
• Adjustment rules: Selecting the **top 5%** (around 25 stocks) of stocks with equal weights allocation based on signal ranking, strictly calculating turnover rate to control costs.

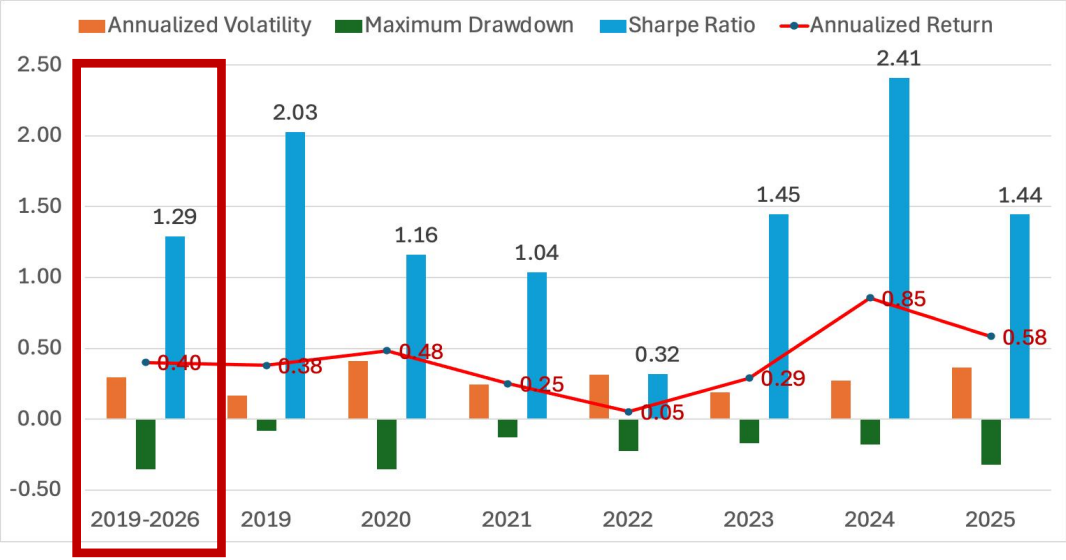
04.

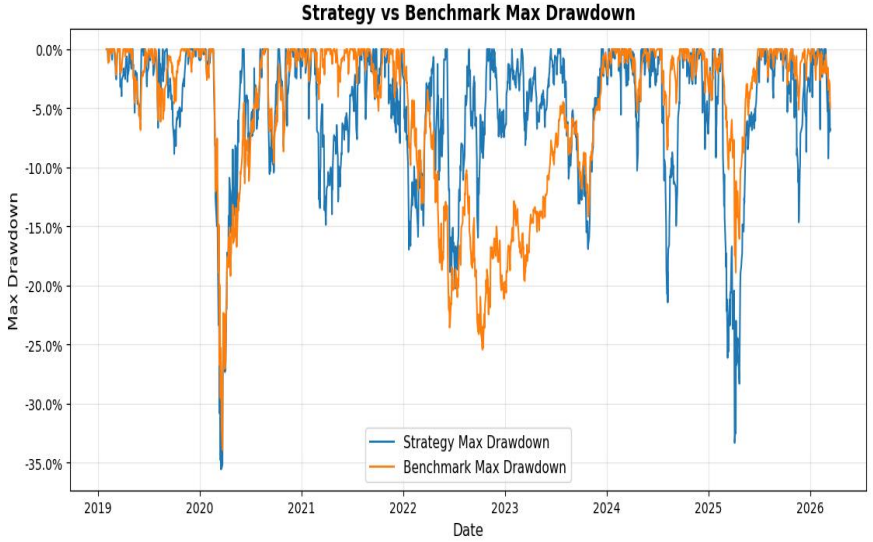
Performance Evaluation

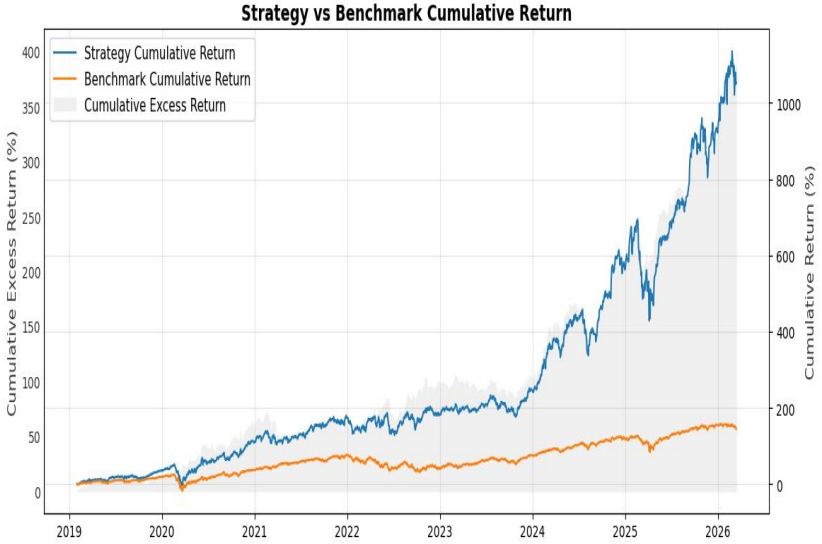
Back-testing a multi-dimensional momentum strategy using S&P 500 daily OHLCV data

Performance Evaluation







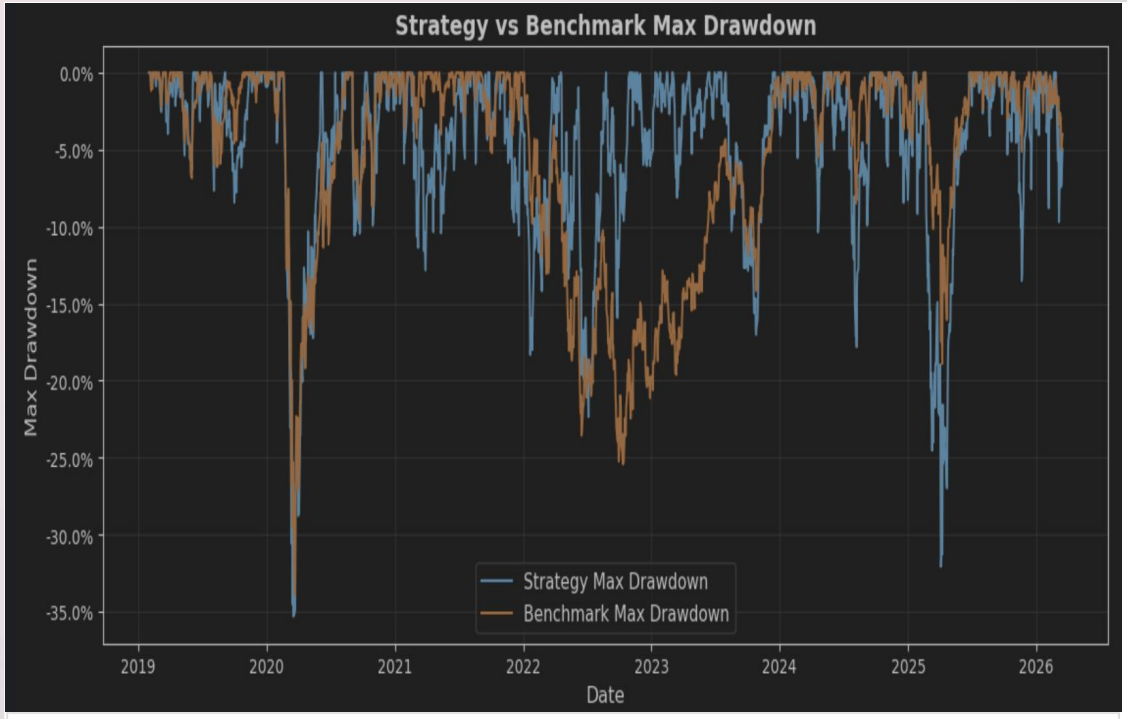


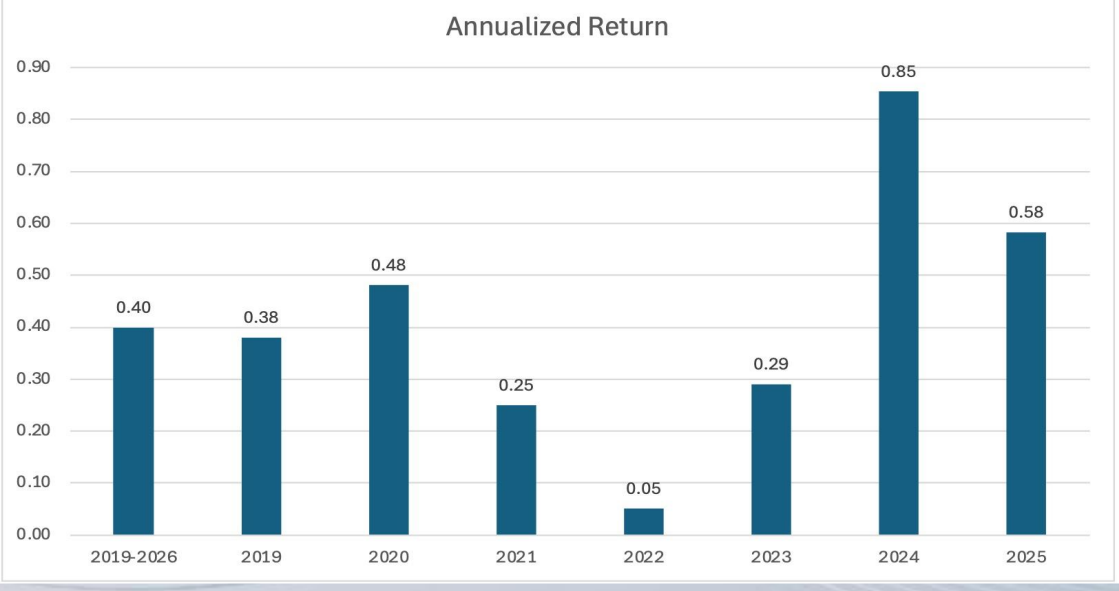
	Strategy	Benchmark
Annualized Return（%）	40	12
Annualized Volatility（%）	29	20
Maximum Drawdown（%）	-35	-37
Sharpe Ratio	1.29	0.69
Sortino Ratio	1.67	
Calmar Ratio	1.13	
Excess Annualized Return（%）	27	
Excess Maximum Drawdown（%）	-17	
Information Ratio (IR)	1.38	
Average Turnover Rate（%）	42	
Average IC Value	0.04	

05.

Risk Analysis & Failure Explanation

The strategy demonstrates robust recovery power but is susceptible to 'Momentum Crashes' during black-swan events (Feb 2020) and rapid macro-regime shifts (2022). These drawdowns are primarily due to the 252-day lookback period's inherent time-lag, which prevents the model from pivoting instantly during sudden market reversals





Case 1: Q1 2020 (The COVID Crash)

Performance: Max Drawdown ~35% (Exceeded Benchmark).

Momentum Lag: 252-day window was too slow for the "V-shape" flash crash.

Liquidity Shock: Vol_Trend failed as panic triggered a universal, non-selective sell-off.

Case 2: 2022 H2 – 2023 Q1 (Rate Hike Cycle)

Performance: Prolonged underperformance during the "Bear Market."

Style Rotation: Rising rates caused a massive shift from Growth (our tilt) to Value.

Whipsaw Risk: Frequent trend reversals led to "paper cut" losses in a directionless market.

Case 1: 2022 H2 – 2023 Q1 (The Monetary Tightening Regime)

Performance Observation:
A period of "flat" annualized returns (+5% in 2022) and a slower recovery compared to the benchmark.

Style Rotation:
Rapid interest rate hikes caused a massive market rotation from Growth (the strategy's natural tilt) to Value; the factors struggled to capture Alpha as high-momentum tech stocks were penalized.

Whipsaw Risk:
Frequent trend reversals during bear market rallies caused "paper cut" losses, as momentum signals were triggered just as temporary rebounds faded.

THANK YOU